JOB BULLETIN
JB-08-0002
Rev 7, Nov 07, 2025
External

Component Tagging Procedure

1.0 SCOPE

This job bulletin describes the numbering convention for components such as fixed equipment, instruments, lines, valves, specialty items, cable trays and cables on the Lilly LP1 project.

2.0 PURPOSE

The purpose of this job bulletin is to document how all equipment, instruments, lines, valves, specialty items, cable trays, and cables should be numbered. This job bulletin will identify the different types of items utilized on this project and provide guidance on numbering.

3.0 APPLICATION

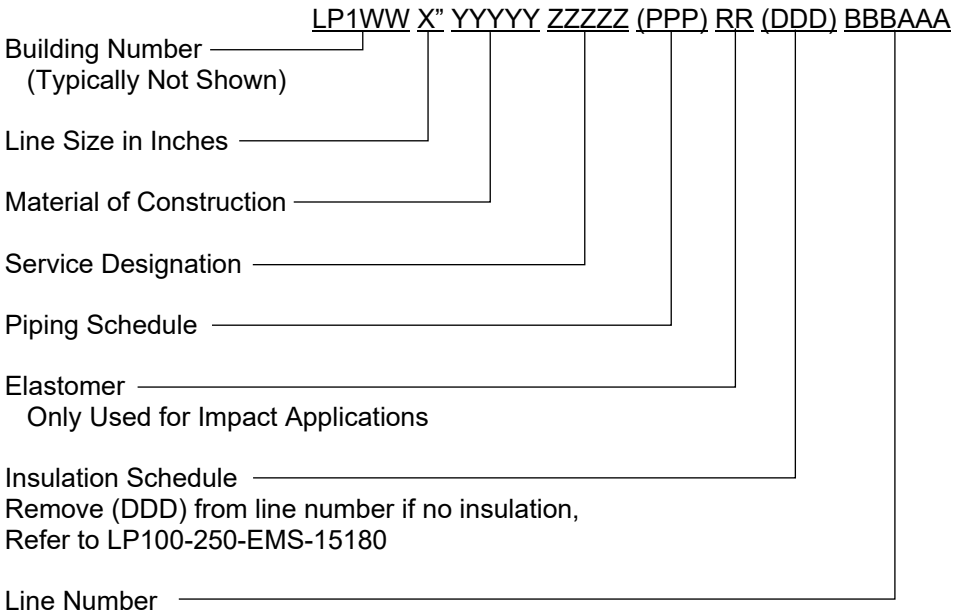
The conventions defined in this job bulletin apply to the items listed in the sections below.

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4.0 DOCUMENT PROCEDURES

4.1 Line Numbers

The following convention shall be used to assign line numbers. Line numbers are unique based on Building Number (LP1WW) and Line Number which includes a P&ID number (BBB or BBBB) and a Sequence Number (AAA). Please review references for lookup tables.



BBB – Last three digits of originating P&ID number (BBBB - Last four digits for HVAC P&IDs)
AAA – Sequential three-digit number

NOTE: Consult Process Engineering Lead Sheet LP100-DI-00000002 for the current, project-approved Materials of Construction, Elastomers, and Insulation Schedules. The line number originating from or going to a piperack P&ID will have a sequence number using the last three digits of the piperack P&ID starting from the Building/Tank Farm break.

References:

- Building Numbers - Refer to JB-08-0001.
- Service Designation - Refer to Process Engineering Lead Sheet LP100-DI-00000002 and Piping Service Index (Attachment D in LP100-250-EMS-15100 Basic Piping Materials and Methods)

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Material of Construction, YYYYY	Material
AL6XN	AL-6XN
CI	CAST IRON
CO	COPPER
CS	CARBON STEEL
DI	DUCTILE IRON
DUP	DUPLEX
GALVS	GALVANIZED STEEL
HAST	HASTELLOY / ALLOY 22
NM	NONMETALLIC
PP	PP / POLYPRO LINED
PVDF	PVDF / KYNAR LINED
SS	STAINLESS STEEL
TFE	TEFLON LINED

Elastomer, RR	Elastomer Material
E	EPDM
K	KALREZ (FFKM)
P	PTFE
PE	PTFE OR EPDM
PK	PTFE OR KALREZ (FFKM)
S	PLATINUM CURED SILICONE
V	VITON

INSULATION SCHEDULE, DDD - Reference LP100-250-EMS-15180 Insulation

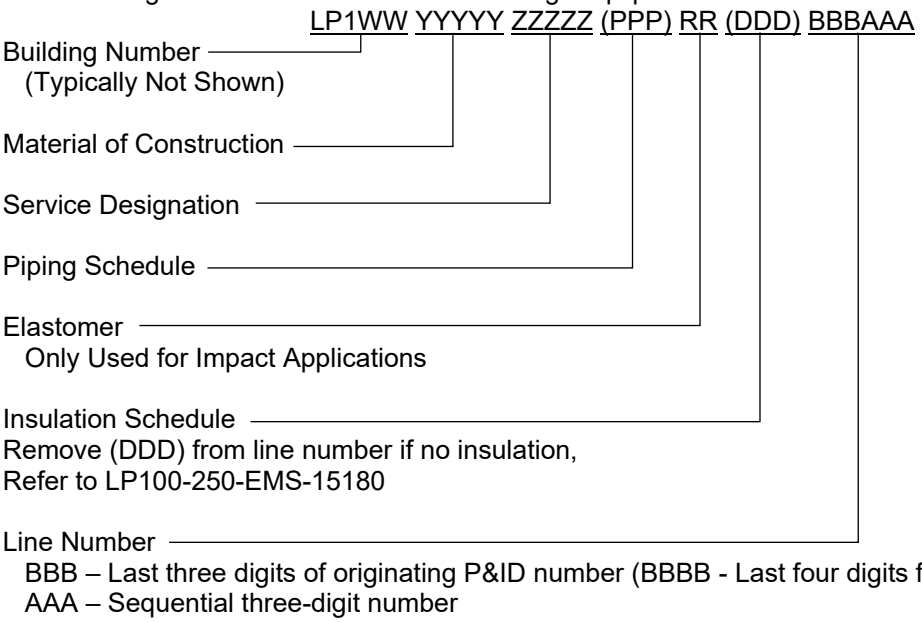
INSULATION SCHEDULE, DDD	OPERATING TEMPERATURE °F	INSULATION MATERIALS
108	-300 TO 0	PIR
107 (INDOOR)	0 TO 125	ELASTOMERIC
108 (OUTDOOR)		PIR
101	125 TO 300	FIBERGLASS
101 (INDOOR)	300 TO 600	FIBERGLASS
103 (OUTDOOR)		MINERAL WOOL
111 (NOTE 1)	-450 TO 600	CELLULAR GLASS

Note 1: For use on Syltherm systems only, as indicated on P&IDs.

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4.2 Equipment Trim Numbers

The following convention shall be used to assign equipment trim numbers.



BBB – Last three digits of originating P&ID number (BBBB - Last four digits for HVAC P&IDs)
AAA – Sequential three-digit number

NOTE: Consult Process Engineering Lead Sheet LP100-DI-00000002 for the current, project-approved Materials of Construction, Elastomers, and Insulation Schedules.

References:

- Building Numbers - Refer to JB-08-0001.
- Service Designation - Refer to Process Engineering Lead Sheet LP100-DI-00000002 and Piping Service Index (Attachment D in LP100-250-EMS-15100 Basic Piping Materials and Methods)

Material of Construction, YYYYY	Material
AL6XN	AL-6XN
CI	CAST IRON
CO	COPPER
CS	CARBON STEEL
DI	DUCTILE IRON
DUP	DUPLEX
GALVS	GALVANIZED STEEL
HAST	HASTELLOY / ALLOY 22
NM	NONMETALLIC
PP	PP / POLYPRO LINED
PVDF	PVDF / KYNAR LINED
SS	STAINLESS STEEL
TFE	TEFLON LINED

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Elastomer, RR	Elastomer Material
E	EPDM
K	KALREZ (FFKM)
P	PTFE
PE	PTFE OR EPDM
PK	PTFE OR KALREZ (FFKM)
S	PLATINUM CURED SILICONE
V	VITON

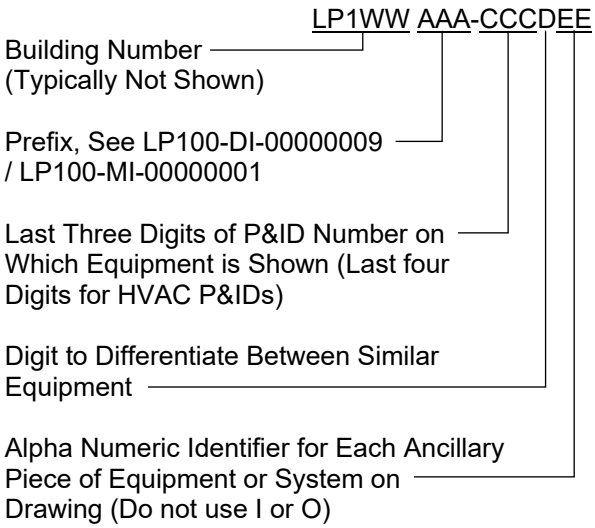
INSULATION SCHEDULE, DDD - Reference LP100-250-EMS-15180 Insulation

INSULATION SCHEDULE, DDD	OPERATING TEMPERATURE °F	INSULATION MATERIALS
108	-300 TO 0	PIR
107 (INDOOR)	0 TO 125	ELASTOMERIC
108 (OUTDOOR)		PIR
101	125 TO 300	FIBERGLASS
101 (INDOOR)	300 TO 600	FIBERGLASS
103 (OUTDOOR)		MINERAL WOOL
111 (NOTE 1)	-450 TO 600	CELLULAR GLASS

Note 1: For use on Syltherm systems only, as indicated on P&IDs.

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4.3 Fixed Process, Utility, and HVAC Equipment Numbers



NOTE: Consult Process Engineering Lead Sheet LP100-DI-00000005 and LP100-DI-0000009 for the current, project-approved Equipment/Component Suffixes and Equipment Prefixes. For HVAC Equipment Prefixes refer to LP100-MI-00000001. For Plumbing equipment, refer to Plumbing Lead Sheet LP1WW-MJ-00000701 for current project-approved Prefixes for the particular building. Plumbing equipment that is not included in a P&ID will drop the P&ID designation on the tag and only use LP1WW AAA-DEE.

Where more than 10 unique equipment identifiers are needed for similar equipment, D may used to indicate the floor location of the equipment and EE may be used with sequential alphabetic identifier. Example: suffix 1A would be used for the first floor first item while suffix 2B would be used for the second floor second item.

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EE Equipment suffixes as shown in table 4.3.1 are recommended for use for the first E digit to convey the service that the component is associated with. This is provided as a guide that may be implemented at the discretion of the Lilly node design team.

Suffix	Typical Association
A	Vessel Outlet (Bottom Valve) Vessel Level (Transmitter and Switches) Vessel Temperature Vessel Pressure (Transmitter and Switches) Vessel Transfer/Recirc Pump and Speed Controller Items associated with primary vessel boundary
B	Bases
C	CIP Lines to vessels
D	Column distillate
E	High Purity Water
EMR	Emergency Relief Header
F	Column Feed Lines or raw material charge
G	Acid
H	Utility Air Lines
I	Not Used
J	Heating/Cooling Medium at equipment (inlet and outlet medium 1: cold)
K	Heating/Cooling Medium at equipment (inlet and outlet medium 2: hot)
L	Open
M	Open
N	Nitrogen
O	Not Used
P	Product Lines
Q	Sampling
R	Recirculating/Reflux Lines
S	Solvent
T	Open
U	Steam/Condensate
V	Vent/Vapor Lines (could be line going to scrubber, vacuum, etc – add second letter)
W	Waste Solvent 2nd character: P = Primary (e.g., WP) S = Secondary (e.g., WS)
X	Spent Solvent
Y	Water Waste Line
Z	Open

Table 4.3.1 EE Component Suffixes

References

- Building Numbers - Refer to JB-08-0001.

Equipment tags shall be reviewed and approved by client engineering team.

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4.4 Electrical Systems Equipment Numbers

Equipment Numbering will be as follows:

31-ELI/2N-2G-01

Building Node:
For Building Number: Refer to JB-08-0001

Sequence Number 01, 02…

Equipment Type:

Column Grid Lines

Type	Description
ASD	ADJUSTABLE SPEED DRIVE
BD	BUS DUCT
DS	DISCONNECT SWITCH
EHT	ELECTRIC HEAT TRACING
ELI	EMERGENCY LIGHTING INVERTER
EMCC	EMERGENCY MOTOR CONTROL CENTER
FOP	FIBER OPTIC PATCH PANEL
GEN	STANDBY GENERATOR
MBP	MAINTENANCE BYPASS PANEL
MCC	MOTOR CONTROL CENTER
MCP	MOTOR CONTROL PANEL
MDP	MAIN DISTRIBUTION PANEL
MSB	MAIN SERVICE BOARD
LC	OUTDOOR LIGHTING CONTACTOR PANEL
LP	480Y/277V LIGHTING PANEL
PDP	480V POWER DISTRIBUTION PANEL
RP	208Y/120V RECEPTACLE PANEL
SWBD	SWITCHBOARD
SWG	SWITCHGEAR
TX	TRANSFORMER
UDP	UNINTERRUPTIBLE POWER DISTR. BOARD
UPP	UNINTERRUPTIBLE POWER SUPPLY PANEL
UPS	UNINTERRUPTIBLE POWER SUPPLY
US	UNIT SUBSTATION
UTX	UNINTERRUPTIBLE TRANSFORMER

Floor Level:

IDENTIFIER	DESCRIPTION
0	Outside or General
1	1st Floor
2	2nd Floor
3	3rd Floor
4	4th Floor
5	5th Floor
6	6th Floor

Floor Level Subtypes

IDENTIFIER	DESCRIPTION
0	GENERAL
B	BASEMENT
M	MEZZANINE
N	INTERSTITIAL
P	PENTHOUSE
R	ROOF
U	UNDERGROUND

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Note: Consult Electrical Legend and Symbols Sheet LP100-EJ-00000002 for current, project-approved Electrical Equipment tagging.

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4.5 Auxiliary Systems Equipment Numbers

The below conventions are to be followed for the following Aux system equipment:

Telecommunication Racks and Cabinets (in IDF Rooms)

TR-LP1-AA-BB-C-D

TR – Telecom Room
LP1 – Project Designator
Building Number: Refer to JB-08-0001
BB – Floor Number
C – Room Designator (A, B, C, D, etc)
D – Rack Number (1, 2, 3, 4, etc)

Telecommunication Racks and Cabinets (in MDFs/Data Centers)

TR-LP1-AA-BB-C-D

TR – Telecom Room
LP1 – Project Designator
Building Number: Refer to JB-08-0001
FR - Frame Room for MDFs or
DC - Data Center
C – Room Designator (A, B, C, D, etc)
D – Rack Number (1, 2, 3, 4, etc)

Telecommunication Racks and Cabinets (Stand Alone Remote Cabinets)

TC-LP1-AA-BB-C

TC – Telecom Cabinet
LP1 – Project Designator
Building Number: Refer to JB-08-0001
BB – Floor Number
C – Cabinet/Rack Designator (A, B, C, D, etc)

Security System Panels (Access Control System) and Fire Alarm System Panels

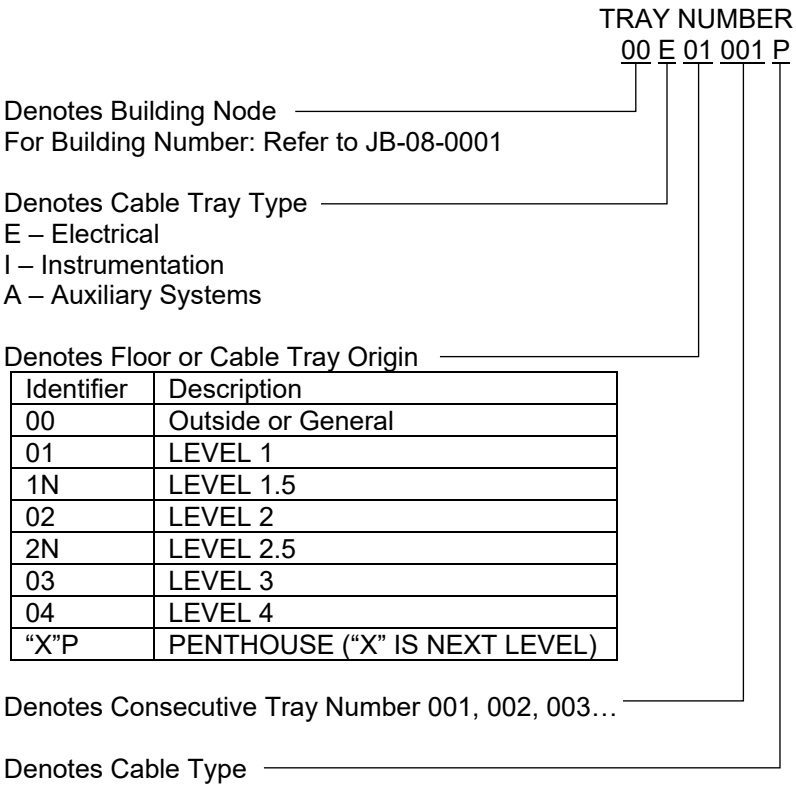
LP1-AA-BB-XXX-D

LP1 – Project Designator
Building Number: Refer to JB-08-0001
BB – Floor Number
ACS – Access Control System or
FAS – Fire Alarm System
D – Panel Designator (A, B, C, D, etc)

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4.6 Electrical and Auxiliary Systems Cable Tray Numbers

Cable Tray Numbers shall be numbered as follows:



Identifier	Description
00	Outside or General
01	LEVEL 1
1N	LEVEL 1.5
02	LEVEL 2
2N	LEVEL 2.5
03	LEVEL 3
04	LEVEL 4
“X”P	PENTHOUSE (“X” IS NEXT LEVEL)

For Electrical:
C – Control
E – Emergency Power
M5 – 5kV Power
M15 – 15kV Power
P – Normal Power
S – Standby Power
U – UPS Power

For Instrumentation Systems:
I – Signal (Cables between instrument / device and SJB, for example: 4-20mA, ON-OFF, 24V DC)
IS – Intrinsically Safe

For Auxiliary Systems:
A – Auxiliary Systems

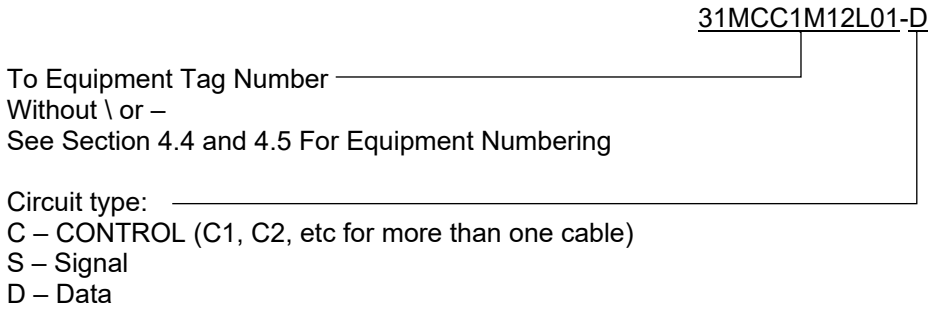
Note: Consult Electrical Legend and Symbols Sheet LP100-EJ-00000002 for current, project-approved Cable Tray tagging.

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4.7 Electrical and Auxiliary Systems Cable Numbers

4.7.1 Auxiliary Systems Cable Numbers

See below examples for numbering Auxiliary Systems cables.





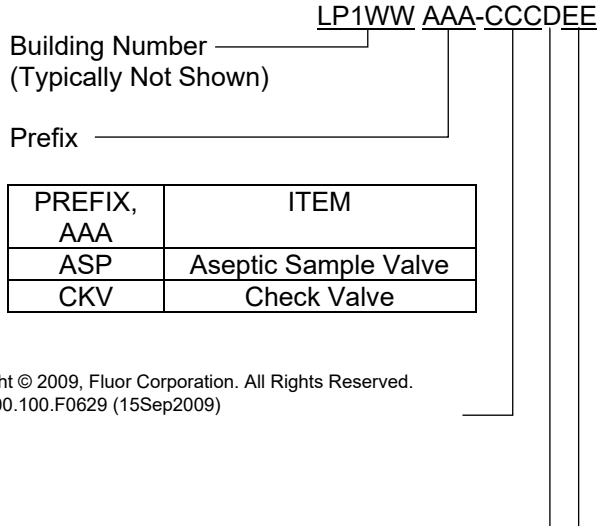
Note:
* Suffix D used to tag Network / Communication cables
(for example: Network cables from LCP to PLC, PLC to PLC, ASD to MCC)

4.7.2 Electrical System Cable Numbering

See project drawings LB100-EJ-00000001, -02, and -10 for detailed electrical numbering requirements and examples.

4.8 Manual Valve Numbers

The following convention shall be used to assign manual valve numbers.



PREFIX, AAA	ITEM
ASP	Aseptic Sample Valve
CKV	Check Valve

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HV	Hand Valve
SAM	Sample Valve, Process

Last Three Digits of P&ID Number on
Which Valve is Shown (Last Four Digits
For HVAC P&IDs)

Digit to Differentiate Between Similar
Equipment

Alpha Numeric Identifier for
Each Ancillary Piece of Equipment or System on
Drawing (Do not use I or O)

NOTE: Consult Process Engineering Lead Sheet LP100-DI-00000003 for the current, project approved Manual Valve Numbering

EE Manual Valve Suffixes as shown in table 4.3.1 are recommended for use for the first E digit to convey the service that the component is associated with. The second digit conveys the sequential count, with a third or fourth digit being used wherever the count exceeds 9 or 99, respectively. This is provided as a guide that may be implemented at the discretion of the Lilly node design team.

References:

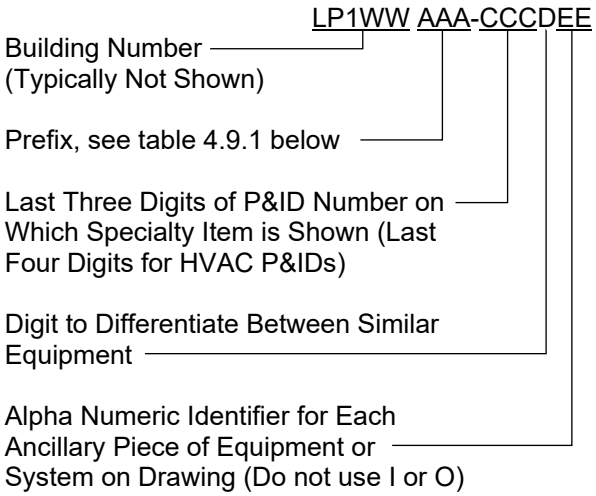
- Building Numbers - Refer to JB-08-0001.

Manual Valve tags shall be reviewed and approved by client engineering team.

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4.9 Piping Specialty Items

The following convention shall be used to assign piping specialty items.



EE Piping Specialty Item suffixes as shown in table 4.3.1 are recommended for use for the first E digit to convey the service that the component is associated with. The second digit conveys the sequential count with a third or fourth digit being used wherever the count exceeds 9 or 99, respectively. This is provided as a guide that may be implemented at the discretion of the Lilly node design team.

References:

- Building Numbers - Refer to JB-08-0001.
- Item Prefix – Refer to Table *4.9.1* – Prefix

Piping Specialty Items tags shall be reviewed and approved by client engineering team.
NOTE: Consult Process Engineering Lead Sheet LP100-DI-00000003 for the current project-approved prefixes for piping specialty items. Plumbing specialty items have also been included in the below table. See notes in below section for specific tagging instructions for Indoor and Outdoor Eyewash and Safety Showers.

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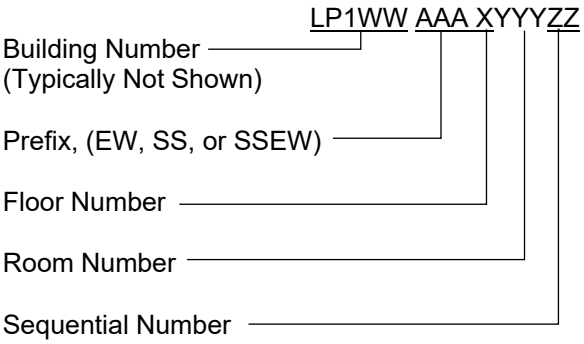
PREFIX	ITEM
AAV	Automatic Air Vent
ACD	Automatic Condensate Drain
ASP	Aseptic Sample Point
BA	Catch Basin
BAS	Breathing Air Station
BFP	Backflow Preventer
BS	Bird Screen
BV	Balancing Valve
CCU	Camlock Connection
CD	Cleaning Device
CFP	Chemical Feed Pot
CPT	Condensate Pump Trap
CTD	Closed Trench Drain (pop up)
DA	Detonation Arrestor
DP	Drip Pan
DPE	Drip Pan Elbow
DT	Dip Tube
EI	Engineered Item
EJ	Expansion Joint
EW	Eyewash Station
EXH	Exhaust Head
FD	Open Floor Drain
FLA	Flame Arrestor
FLC	Flexible Connector
FLX	Flexible Hose
FUN	Funnel
HD	Hub Drain
HST	Hose Station
LD	Liquid Drainer
LPCV	Low Pressure Check Valve
LUB	Lubricator
MV	Mixing Valve
MXS	Mixing Station
OTD	Open Trench Drain
PBV	Passive Butterfly Valve
PC	Pipe Cover
PD	Popup Drain
QCU	Quick Connect Unit
RS	Removable Spool
RSPTR	Respirator
SAC	Small Additional Container
SAM	Sample Valve
SBV	Split Butterfly Valve
SD	Suction Diffuser
SEP	Separator
SG	Sight Glass

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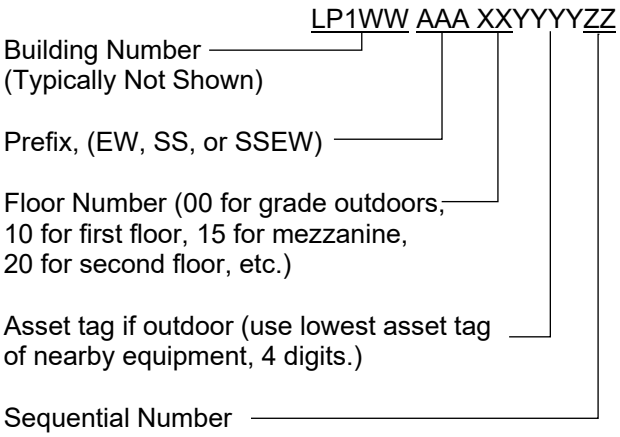
PREFIX	ITEM
SLR	Silencer
SLP	Seal Pot
SLV	Spring Loaded Valve
SPA	Sparger
SPEC	Spectacle Blind
SS	Safety Shower
SSEW	Safety Shower and Eyewash Combination
STR	Strainer
STT	Steam Trap
TK	Tank (Expansion)
WH	Water Heater
WHA	Water Hammer Arrestor

Table 4.9.1 – Prefix is per Process P&ID Lead Sheets LP100-DI-00000003.

For, indoor Eyewash (EW), Safety Shower (SS) or Safety Shower/Eyewash Combination (SSEW) the following convention shall be used:

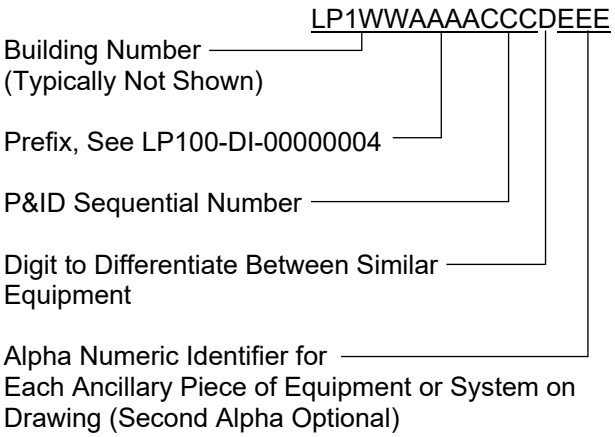


For, outdoor Eyewash (EW), Safety Shower (SS) or Safety Shower/Eyewash Combination (SSEW) the following convention shall be used:



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4.10 Instrument Numbers



EEE Instrument Numbers suffixes as shown in table 4.3.1 are recommended for use for the first E digit to convey the service that the component is associated with. The second and third digits are optional and should only be used where there are more than one (1) of the same instrument type. This is provided as a guide that may be implemented at the discretion of the Lilly node design team.

For limit switches associated with Eyewash (EW), Safety Shower (SS) or Safety Shower/Eyewash Combinations (SSEW), the instrument tagging will not follow the CCC P&ID sequence number and D digit to differentiate between similar instruments but instead will be tagged using the floor number, room number, and sequential number (in the case of Indoor SSEW) or floor number, asset tag from nearby equipment, and sequential number (in the case of Outdoor SSEW). The suffix will continue to include an alpha numeric identifier.

For gas detectors located inside manufacturing buildings, the instrument tagging will use room numbers in place of CCC P&ID sequence numbers.

References:

- Building Numbers - Refer to JB-08-0001.

Instrument Numbers shall be reviewed and approved by client engineering team.

4.11 Instrument Cable Numbers

Cable numbers shall contain FROM/TO information, indicating where the cable comes “FROM” and where it goes “TO”. For further details on the cable numbering, refer to below specification sections.

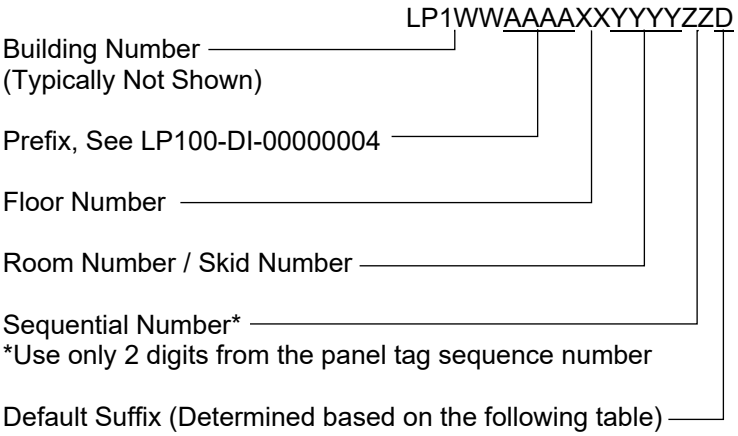
References:

- See Section 3.02.D.9.G of LP100-270-EMS-16920 INSTRUMENT SIGNAL AND POWER WIRING.

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4.12 Miscellaneous Delta V Panel Monitoring Tags

Tagging for DeltaV panel power supply monitoring, Temperature monitoring, and Miscellaneous I/O to follow the below specification sections.



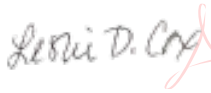
Panel Type	Default Suffix	Temp High	Temp Low
IO	A		
IPP	B	H	L
CTL	C		
NER	D		
SER	E		

References:

- See Section 1.09 of LP100-270-EMS-16981 CONTROL PANELS for more information about Panel Types and Panel Numbering

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Leslie Cox
Engineering Director

[LC]:[LNP/MJR]:

Attachment(s)

- None

References(s)

- JB-08-0001 Drawing and Document Numbering Procedure
- LP100-250-EMS-15100 Basic Piping Materials and Methods
- LP100-250-EMS-15113 Piping Specialty Items
- LP100-250-EMS-15180 Insulation
- LP100-270-EMS-16981 CONTROL PANELS
- LP100-270-EMS-16920 INSTRUMENT SIGNAL AND POWER WIRING.
- LP100-DI-00000001 through 9 Process Engineering Lead Sheets
- LP100-EJ-00000002 Electrical Legend and Symbols Sheet
- LP1WW-MJ-00000701 Plumbing Lead Sheet (for Building Number WW)
- LP100-MI-00000001 HVAC AIRFLOW & INSTRUMENTATION DIAGRAM LEAD SHEET
- LP100-MI-00000002 HVAC AIRFLOW & INSTRUMENTATION DIAGRAM LEAD SHEET

File No.: 100A-504-RL

Change Log

Rev. No.	Date	Description / Changes Incorporated
0	Feb 27, 2023	Initial Revision with holds on Electrical Equipment, Cabling and Instrument Cabling
1	June 1, 2023	Updated tagging for equipment to allow unique tagging of more than 10 similar equipment. Included tagging methodology for electrical equipment and electrical cable tray. Revised to include component suffix guidance for Equipment, Piping Specialty Items, Manual Valves, and Instruments. Revised references to project issued EMS specifications.

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Rev. No.	Date	Description / Changes Incorporated
2	Oct 12, 2023	Renamed job bulletin to Component Tagging Procedure and updated scope and purpose to include valves, specialty items, cables and cable tray. Removed requirement for instrument number suffix to be two digits and noted second digit as optional. Added HVAC P&ID specific component tagging with four digits from the P&ID number. Updated table 4.3.1 of EE component suffixes. Revised electrical equipment numbering to start with building number then equipment type. Updated Cable Tray numbering type conventions and included Auxiliary Systems tray numbering. Updated valve prefix and suffix numbering. Updated specialty item prefix numbering to include plumbing specialties in Table 4.8.1 and updated suffix numbering. Included cable numbering and references to instrumentation cable numbering procedure in project specifications. Updated references to remove LP100-EMS-01910 and instead referring to P&ID Lead Sheets. Updated snapshots from Lead Sheets to reflect latest master markups.
3	Mar 19, 2024	Updated the cable tray numbering methodology. Updated tagging for piping specialty items to clarify eyewash / safety shower tagging based on room number for indoor showers and nearby asset tags for outdoor items. Included eyewash / safety shower instrument tagging methodology. Updated piping specialty item codes to include additional item types. Included note on line numbering for lines at the building/Tank Farm interface point. Included note that service designations can be referred to on the process master PID lead sheet. Updated tables to include additional insulation and material of construction codes. Updated instrument numbering to allow suffix to have third optional alpha suffix. Updated Auxiliary equipment numbering convention and moved Aux equipment to its own section 4.5 and updated subsequent section numbers.
4	Aug 30, 2024	Updated electrical equipment type list in section 4.4. Updated cable tray numbering in section 4.6. Added reference to electrical legend and symbols sheet LP100-EJ-00000002. Updated section 4.10 instrument numbering to remove spaces between fields. Updated JB to be external.
5	Apr 1, 2025	Updated section 4.2 to note that elastomer code should be used only for impact applications (not just tri-clamp applications). Added exception in section 4.10 for manufacturing building gas detector tagging to use room number in lieu of PID number. Added section 4.12 for Misc DeltaV Panel Monitoring tagging. Added reference to two HVAC AF&ID Lead Sheets and HVAC equipment prefix reference lead sheet in section 4.3. Added PBV – Passive Butterfly Valve prefix for piping specialty item numbers in section 4.9.
6	June 19, 2025	Updated section 4.7 to add note about use of suffix D for tagging network cables. Updated section 4.12 Misc DeltaV Panel Monitoring tagging to satisfy the needed 4 digits for room / skid numbers.
7	Nov 7, 2025	Updated section 4.7 with a note that Electrical Cable Numbering is per listed project drawings.

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